

Solving systems of eqn

$$\begin{array}{rcl} a+b=10 & \longrightarrow & a=8 \\ a+2b=12 & & b=2 \\ \hline -b=-2 & & \\ \hline \boxed{b=2} & \text{into } a+b=10 & \\ & \hline & \boxed{a=8} \end{array}$$

Ex →

$$\begin{array}{rcl} 5a+b=12 & \div 5 & a=2.4 \\ 4a-3b=6 & \div 4 & b=2 \\ \hline a+0.2b=3.4 & & \\ a+0.25b=1.5 & & \\ \hline 0.05b=1.9 & & \\ \hline \boxed{b=38} & & \end{array}$$

Ex →

$$\begin{array}{rcl} 5a+b=12 & \div 5 & a+0.2b=2.4 \\ 3b=6 & \div 3 & b=2 \\ \hline b=2 & & \end{array}$$

Ex →

$$\begin{array}{rcl} 2a+5b=46 & \div 2 & a+2.5b=23 \\ 4 \times (8a+b=32) & \div 8 & a+0.125b=4 \\ \hline 8a+20b=184 & & \\ 8a+b=32 & & \\ \hline 19b=152 & & \\ \hline b=8 & & \end{array}$$

$$a = \frac{24}{8}$$

$$\boxed{a=3}$$

$$\begin{array}{rcl} a+2.5b=23 \\ a+0.125b=4 \\ \hline 2.375b=19 \\ \hline b=8 \\ \hline \boxed{b=8} \quad \boxed{a=3} \end{array}$$

$$\begin{array}{rcl} a+2.5b=23 \\ a+0.125b=4 \\ \hline 2.375b=19 \\ \hline b=8 \end{array}$$

$$\begin{array}{r} 2500 \ 49 \\ 0.125 \\ \hline 2.375 \end{array}$$

What if the system is singular (redundant)?

divide by coefficient of a

$$a+b=10$$

$$\rightarrow a+b=10$$

$$2a+2b=20$$

$$\rightarrow a+b=10$$

$$\begin{array}{r} - \\ - \\ - \\ \hline 0 \ 0 \ =0 \end{array}$$

$\hookrightarrow$ not giving any info.

Solved System

$$a=x$$

$$b=10-x$$

What if the system is singular (contradictory)?

$$a+b=10$$

divide by coefficient of a
 $\rightarrow a+b=10$

$$2a+2b=24$$

$$\rightarrow a+b=12$$

$$\begin{array}{r} - \\ - \\ - \\ \hline 0 \ =2 \end{array}$$

$\hookrightarrow$ contradiction

Solved system

$\rightarrow$ no solⁿ

$$\text{Ex} \rightarrow 5a+b=11$$

$$10a+2b=22$$

$$\rightarrow a+0.2b=2.2$$

$$\rightarrow a+0.2b=2.2$$

$$\begin{array}{r} - \\ - \\ - \\ \hline 0 \ =0.3 \end{array}$$

$\hookrightarrow$ contradiction

$\hookrightarrow$ no solⁿ

Solve system

no solⁿ

→ Solving system of eqⁿ with more variables

Elimination method

Divide each row by coefficient of a

$$\begin{array}{lcl} a + b + 2c = 12 & \longrightarrow & a + b + 2c = 12 \quad - \textcircled{I} \\ 3a - 3b - c = 3 & \longrightarrow & a - b - \frac{c}{3} = 1 \quad - \textcircled{II} \\ 2a - b + 6c = 24 & \longrightarrow & a - \frac{b}{2} + 3c = 12 \quad - \textcircled{III} \end{array}$$

$$\textcircled{II} - \textcircled{I}$$

$$\frac{3b}{2} - c = 0$$

$$\textcircled{II} - \textcircled{III}$$

$$2b + \frac{7c}{3} = 11$$

resulting system

$$a + b + 2c = 12$$

$$2b + \frac{7c}{3} = 11 \quad \rightarrow \quad b + \frac{7}{6}c = 11$$

$$\frac{3b}{2} - c = 0 \quad \rightarrow \quad b - \frac{2}{3}c = 0$$

$$\begin{array}{r} b - \frac{2}{3}c = 0 \\ - \quad \frac{7}{6}c = 11 \\ \hline \frac{1}{6}c = 11 \end{array}$$

$$\frac{3b}{2} - 6 = 0$$

$$\frac{3b}{2} = 6$$

$$\boxed{b = 4}$$

$$\boxed{c = 6}$$

$$a + 4 + 12 = 12$$

$$\boxed{a = -4}$$

matrix row reduction

original

$$5a + b = 17$$

$$4a - 3b = 6$$

matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix}$$

intermediat

$$a + 0.2b = 3.4$$

$$b = 2$$

upper diagonal matrix

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Row echelon form

Solved system

$$a = 3$$

$$b = 4$$

Diagonal matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reduced row echelon form

Ex →

$$a + b = 10$$

$$\rightarrow a + b = 10$$

$$2a + 2b = 20$$

$$0a + 0b = 0$$

original matrix

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

upper diagonal matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

Row echelon form

Ex →

$$5a + b = 11$$

$$\rightarrow a + 0.2b = 2.2$$

$$10a + 2b = 22$$

$$0a + 0b = 0$$

original matrix

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix}$$

upper diagonal matrix

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

ex → original system

$$0a + 0b = 0$$

$$0a + 0b = 0$$

original matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

intermediate system

$$0a + 0b = 0$$

$$0a + 0b = 0$$

upper diagonal matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Row echelon form

Row echelon form

$$\begin{bmatrix} 1 & * & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & * \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & * \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Row operations that preserve singularity

Switching rows

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix}$$

$$\begin{aligned} \det &= 5 \cdot 3 - 1 \cdot 4 \\ &= 15 - 4 \\ &= 11 \end{aligned}$$

↳ non singular matrix

$$\begin{bmatrix} 4 & 3 \\ 5 & 1 \end{bmatrix}$$

$$\begin{aligned} \det &= 4 \cdot 1 - 5 \cdot 3 \\ &= 4 - 15 \\ &= -11 \end{aligned}$$

↳ non singular matrix

Switching row doesn't change singularity

multiplying a row by a (non zero) scalar

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix}$$

$$\begin{aligned} \det &= 15 - 4 \\ &= 11 \end{aligned}$$

$$[5 \ 1] \times 10 = [50 \ 10]$$

$$\begin{bmatrix} 50 & 10 \\ 4 & 3 \end{bmatrix}$$

$$\begin{aligned} \det &= 150 - 40 \\ &= 110 \\ &= 10 \times 11 \end{aligned}$$

Adding a row to another row

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix}$$

$$\det = 15 - 4 = 11$$

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 9 & 4 \\ 8 & 6 \end{bmatrix}$$

$$\det = 54 - 32 = 22 = 2 \times 11$$

adding a row to another row does not change determinants

$$\text{ex} \rightarrow x + y = 4 \quad \times 6 \rightarrow 6x + 6y = 24$$

$$= 6x + 2y = 16$$

$$-6x + 2y = 16$$

$$8y = 40$$

$$\boxed{y = 5}$$

$$x + 5 = 4$$

$$\boxed{x = -1}$$

ex $\rightarrow$

$$A = \begin{bmatrix} 4 & -3 \\ 2 & -8 \end{bmatrix}$$

$$\det(A) = -32 + 6$$

$$= -26$$

$\hookrightarrow$ non-singular

ex $\rightarrow$

$$\begin{bmatrix} -3 & 8 & 1 \\ 2 & 2 & -1 \\ -5 & 6 & 2 \end{bmatrix}$$

$$\begin{aligned} \det &= -3(4 + 6) - 8(4 - 5) + 1(12 + 10) \\ &= -30 + 8 + 22 \\ &= 0 \rightarrow \text{singular} \end{aligned}$$

ex $\rightarrow$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ 2a-d & 2b-e & 2c-f \end{bmatrix}$$

$$\det = a(e(2c-f) - f(2a-d)) - b(d(2b-e) - e(2c-f)) + c(2d(2b-e) - e(2a-d))$$

Rank of a matrix

Systems of information

System 1

The dog is black
The cat is orange

Two sentences

Two pieces of info

Rank = 2

System 2

The dog is black
The dog is black

Two sentences

~~Two~~ one piece of info

Rank = 1

System 3

The dog
The dog

Two sentences

Zero piece of info

Rank = 0

Systems of eqn

$$a + b = 0$$

$$a + 2b = 0$$

Two eqn

Two pieces of info

Rank = 2

$$a + b = 0$$

$$2a + 2b = 0$$

Two eqn

~~Two~~ one piece of info

Rank = 1

$$0a + 0b = 0$$

$$0a + 0b = 0$$

Two eqn

~~Two~~ zero pieces of info

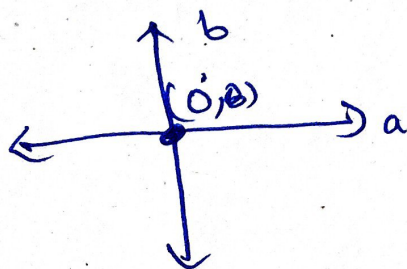
Rank = 0

Rank and soln of the system

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Rank = 2

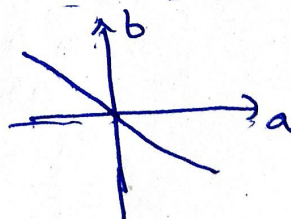
Dimension of soln space = 0



$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Rank = 1

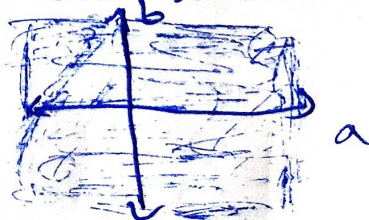
Dimension of soln space = 1



$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank = 0

Dimension of soln space = 2



Rank = 2 - Dimension of soln space

Rank and Singularity

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Rank=2

↓
nonsingular

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Rank=1

↓
singular

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank=0

↓
singular

ex →

$$\begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}$$

Rank=2

ex →

$$\begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$$

Rank=1

Rank of a matrix in general

System 1

$$\begin{aligned} a+b+c &= 0 \\ a+2b+c &= 0 \\ a+b+2c &= 0 \end{aligned}$$

3 eqn

3 Pieces of info

Rank=3

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

System 2

$$\begin{aligned} a+b+c &= 0 \\ a+b+2c &= 0 \\ a+b+3c &= 0 \end{aligned}$$

3 eq

2 Pieces of info

Rank=2

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 3 \end{bmatrix}$$

System 3

$$\begin{aligned} a+b+c &= 0 \\ 2a+2b+2c &= 0 \\ 3a+3b+3c &= 0 \end{aligned}$$

3 eqn

~~2~~ Pieces of info

Rank=0

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$$

System 4

$$\begin{aligned} 0a+0b+0c &= 0 \\ 0a+0b+0c &= 0 \\ 0a+0b+0c &= 0 \end{aligned}$$

3 eq

0 Pieces of info

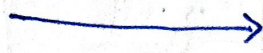
Rank=0

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Row echelon form

original matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix}$$

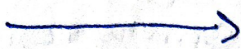


Row echelon form

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Rank 2
2 ones in the diagonal

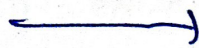
$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

Rank 1
1 one in the diagonal

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$



$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

0 ones in the diagonal
Rank = 0

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0.2 \\ 1 & -0.75 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0.2 \\ 0 & -0.95 \end{bmatrix}$$

Divide each
row by the
left most
coefficient

$$\begin{array}{r} 1 - 0.75 \\ 1 \quad 0.20 \\ \hline 0 \quad -0.95 \end{array}$$

Row echelon form for singular matrices

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 1 & 0.2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 2 & 3 \end{bmatrix}$$

Row echelon form for singular matrices

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Row echelon form in general

$$\begin{aligned}a + b + 2c &= 12 \\ 3a - 3b - c &= 3 \\ 2a - b + 6c &= 24\end{aligned}$$

matrix

$$\begin{bmatrix} 1 & 1 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 6 \end{bmatrix}$$



$$\begin{aligned}a + b + 2c &= 12 \\ -b - 2c &= -33 \\ 6c &= 18\end{aligned}$$

Row echelon form

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -6 & -2 \\ 0 & 0 & 6 \end{bmatrix}$$

Row echelon form

$$\begin{bmatrix} 2 & * & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 3 & * & * \\ 0 & 0 & 0 & -5 & * \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & * & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 4 & * & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

→ Zero rows at bottom

→ Each row has a pivot

(left most non-zero entry)

→ Every pivot is to the right of the pivots on the rows above

Important note

$$\begin{bmatrix} 3 & * & * & * & * \\ 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & -4 & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} &\xrightarrow{\div 3} \\ &\xrightarrow{\div 1} \\ &\xrightarrow{\div (-4)} \end{aligned}$$

$$\begin{bmatrix} 1 & * & * & * & * \\ 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Note! → in general, pivots different than 1 are allowed.

Ex →

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Row echelon form

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Subtract the first row from the second and the third ones

What if the matrix is singular?

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 3 \end{bmatrix}$$

Row echelon form

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

~~First row~~

~~2nd row first row~~

Subtract the first row from the second and the third ones

Subtract twice the second row from the third one

Matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 3 & 3 & 3 \end{bmatrix}$$

Row echelon form

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Subtract twice the first row from the second row

Subtract three times the first row from the third row

Reduced row echelon form

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix} \xrightarrow{\text{upper triangular matrix}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Row echelon form

Diagonal matrix

Reduced row echelon form

Reduced row echelon form

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{array}{r} 0 \quad 1 \\ \times \quad 0.2 \\ \hline 0 \quad 0.2 \end{array}$$

$$\begin{array}{r} 0.2 \\ -0.2 \\ \hline 1 \quad 0 \end{array}$$

General form

$$\begin{bmatrix} \textcircled{1} & 0 & 0 & 0 & 0 \\ 0 & \textcircled{1} & 0 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 0 & 0 \\ 0 & 0 & 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 & 0 & \textcircled{1} \end{bmatrix}$$

$$\begin{bmatrix} \textcircled{1} & * & 0 & 0 & * \\ 0 & 0 & \textcircled{1} & 0 & * \\ 0 & 0 & 0 & \textcircled{1} & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

→ is in row echelon form

→ Each pivot is a 1

→ Any number above a pivot is 0

→ Rank of the matrix is the number of pivots

Ex →

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$2R_2 - R_1$

$$\begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$5R_3 + R_1$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{4R_3 - R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reduced echelon form matrix

Augmented and The Gaussian Elimination Algorithm

Pivot $\rightarrow$
$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 1 \\ 2 & 2 & 4 & -2 \\ 4 & 1 & 0 & -1 \end{array} \right] \xrightarrow{R_1/2} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 2 & 2 & 4 & -2 \\ 4 & 1 & 0 & -1 \end{array} \right] \xrightarrow{\substack{R_2-2R_1 \\ R_3-4R_1}} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 3 & 3 & -3 \\ 0 & 3 & -2 & -3 \end{array} \right]$$

Pivot

Augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right] \xrightarrow{R_2 \times 5} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 5 & 0 \end{array} \right] \xleftarrow{3R_2-R_3} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 3 & -2 & -3 \end{array} \right]$$

$$c=0$$

$$b+c=-1$$

$$\boxed{b=-1}$$

~~$$a+b=c$$~~

$$a - \frac{b}{2} + \frac{c}{2} = \frac{1}{2}$$

$$a - \frac{1}{2} = \frac{1}{2}$$

$$\boxed{a=0}$$

$$R_1 - \frac{1}{2}R_3$$

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & -\frac{1}{2} \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\boxed{a=0, b=-1, c=0}$$

what if the system is singular?

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 2 & 4 & 5 & 1 \\ 3 & 6 & 4 & 6 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 0 & 0 & 7 & 9 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

~~$$0a + 0b + 0c = 0$$~~

$\hookrightarrow$ system has many soln

$$0a + 0b + 0c = 4$$

$\hookrightarrow$ system has no possible soln